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4 Module 4 | Examining Distributions (continued)

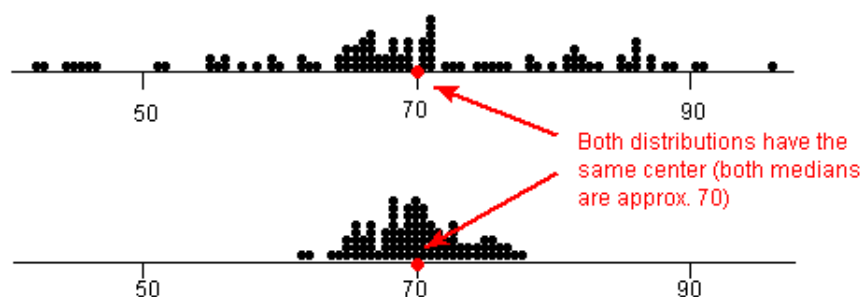
4.4 Numerical Measures

4.4.5 Module 4 | Measures of Spread (1 of 5)

Learning Objectives

- Relate measures of center and spread to the shape of the distribution, and choose the appropriate measures in different contexts.

Introduction So far we have learned about different ways to quantify the center of a distribution. A measure of center by itself is not enough, though, to describe a distribution. Consider the following two distributions of exam scores. Both distributions are centered at 70 (the median of both distributions is approximately 70), but the distributions are quite different. The first distribution has a much larger variability in scores compared to the second one.



In order to describe the distribution, we therefore need to supplement the graphical display not only with a measure of center, but also with a measure of the variability (or spread) of the distribution.

In this section, we will discuss the three most commonly used measures of spread:

- Range
- Inter-quartile range (IQR)
- Standard deviation

Like the different measures of center, these measures provide different ways to quantify the variability of the distribution.

Range The *range* covered by the data is the most intuitive measure of variability. The range is exactly the distance between the smallest data point (min) and the largest one (Max).

- Range = Max - min

Note: When we first looked at the histogram, and tried to get a first feel for the spread of the data, we were actually *approximating* the range, rather than calculating the exact range.

Example (A) Best Actress Oscar Winners

We will continue with the Best Actress Oscar winners example.

34 34 27 37 42 41 36 32 41 33 31 74 33 49 38 61 21 41 26 80 42 29 33 36 45 49 39 34 26 25 33 35 35 28 30
29 61 32 33 45 29 62 22 44

In this example:

- min = 21 (Marlee Matlin for *Children of a Lesser God*, 1986)
- Max = 80 (Jessica Tandy for *Driving Miss Daisy*, 1989)

The range covered by all the data is $80 - 21 = 59$ years.

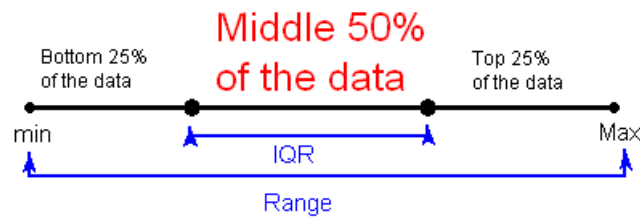
4.4.6 Module 4 | Measures of Spread (2 of 5)

Learning Objectives

- Relate measures of center and spread to the shape of the distribution, and choose the appropriate measures in different contexts.

Inter-Quartile Range (IQR) While the range quantifies the variability by looking at the range covered by *ALL* the data, the IQR measures the variability of a distribution by giving us the range covered by the *MIDDLE 50%* of the data.

The following picture illustrates this idea: (Think about the horizontal line as the data ranging from the min to the Max).

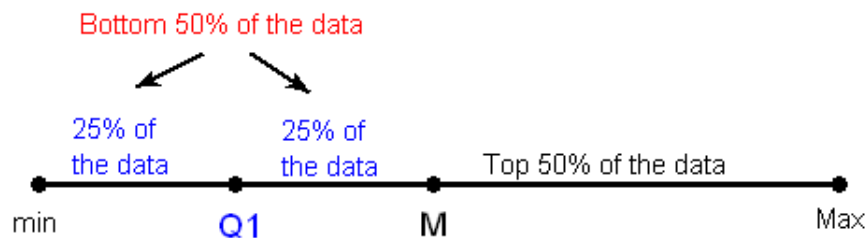


Here is how the IQR is actually found:

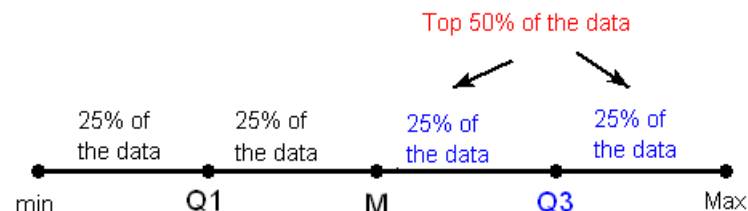
1. Arrange the data in increasing order, and find the median M . Recall that the median divides the data, so that 50% of the data points are below the median, and 50% of the data points are above the median.



2. Find the median of the lower 50% of the data. This is called the first quartile of the distribution, and the point is denoted by $Q1$. Note from the picture that $Q1$ divides the lower 50% of the data into two halves, containing 25% of the data points in each half. $Q1$ is called the first quartile, since one quarter of the data points fall below it.

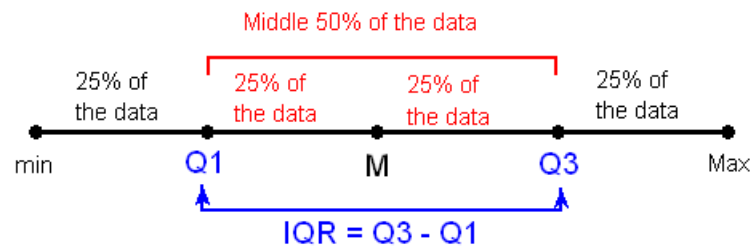


3. Repeat this again for the top 50% of the data. Find the median of the top 50% of the data. This point is called the third quartile of the distribution, and is denoted by $Q3$. Note from the picture that $Q3$ divides the top 50% of the data into two halves, with 25% of the data points in each. $Q3$ is called the third quartile, since three quarters of the data points fall below it.



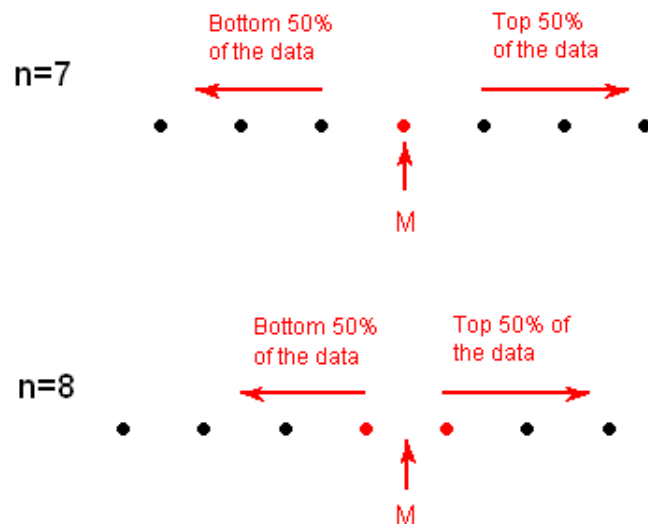
4. The middle 50% of the data falls between $Q1$ and $Q3$, and therefore:

$$IQR = Q3 - Q1$$



Comments:

1. The last picture shows that Q1, M, and Q3 divide the data into four quarters with 25% of the data points in each, where the median is essentially the second quartile. The use of $IQR = Q3 - Q1$ as a measure of spread is therefore particularly appropriate when the median M is used as a measure of center.
2. We can define a bit more precisely what is considered the bottom or top 50% of the data. The bottom (top) 50% of the data is all the observations whose position in the ordered list is to the left (right) of the location of the overall median M . The following picture will visually illustrate this for the simple cases of $n = 7$ and $n = 8$.



Note that when n is odd (as in $n = 7$ above), the median is *not* included in either the bottom or top half of the data; When n is even (as in $n = 8$ above), the data are naturally divided into two halves.

4.4.7 Module 4 | Measures of Spread (3 of 5)

Learning Objectives

- Relate measures of center and spread to the shape of the distribution, and choose the appropriate measures in different contexts.

Example (B) Best Actress Oscar Winners

To find the IQR of the Best Actress Oscar winners distribution, it will be convenient to use the stemplot.

```

2|12
2|56678999          Bottom Half
3|012233333344
3|5566789
4|1112244
4|599
5|
5|
6|112              Top Half
6|
7|4
7|
8|0

```

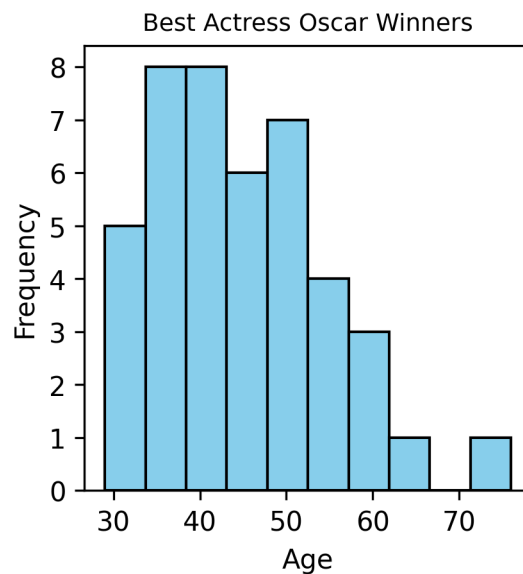
Q1 is the median of the bottom half of the data. Since there are 22 observations in that half, Q1 is the mean of the 11th and 12th ranked observations in that half:

$$Q1 = \frac{30 + 31}{2} = 30.5$$

Similarly, Q3 is the median of the top half of the data, and since there are 22 observations in that half, Q3 is the mean of the 11th and 12th ranked observations in that half:

$$Q3 = \frac{42 + 42}{2} = 42 \quad \text{IQR} = (42 - 30.5) = 11.5 \quad (1)$$

Note that in this example, the range covered by all the ages is 59 years, while the range covered by the middle 50% of the ages is only 11.5 years. While the whole dataset is spread over a range of 59 years, the middle 50% of the data is packed into only 11.5 years. Looking again at the histogram will illustrate this:



Comment Software packages use different formulas to calculate the quartiles Q1 and Q3. This should not worry you, as long as you understand the idea behind these concepts. For example, here are the quartile values provided by three different software packages for the age of best actress Oscar winners:

R:

```

> summary(actress)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  21.00  32.50   35.00   38.53   41.25   80.00

```

Minitab:

Descriptive Statistics: Age

Variable	N	Mean	Median	TrMean	StDev	SE Mean
actress	32	38.53	35.00	36.89	12.95	2.29
Variable	Minimum	Maximum	Q1	Q3		
actress	21.00	80.00	31.50	41.75		

Excel:

Q1	32.5
Q3	41.25

Note that Q1 and Q3 as reported by the various software packages differ from each other and are also slightly different from the ones we found here. There are different acceptable ways to find the median and the quartiles. These can give different results occasionally, especially for datasets where n (the number of observations) is fairly small. As long as you know what the numbers mean, and how to interpret them in context, it doesn't really matter much what method you use to find them, since the differences are really negligible.

4.4.8 Module 4 | Measures of Spread (4 of 5)

Learning Objectives

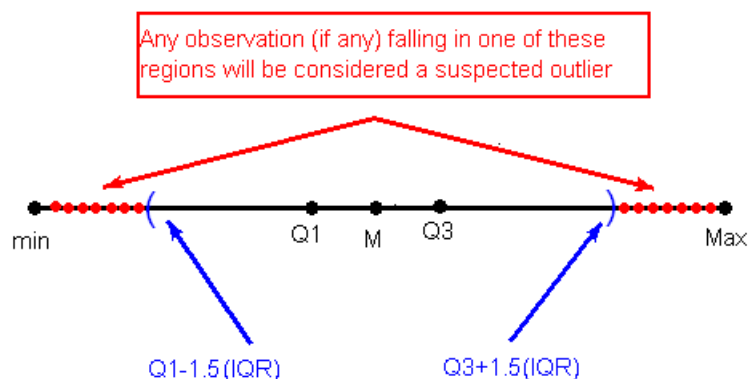
- Summarize and describe the distribution of a quantitative variable in context: a) describe the overall pattern, b) describe striking deviations from the pattern.
- Relate measures of center and spread to the shape of the distribution, and choose the appropriate measures in different contexts.

Using the IQR to Detect Outliers So far we have quantified the idea of center, and we are in the middle of the discussion about measuring spread, but we haven't really talked about a method or rule that will help us classify extreme observations as outliers. The IQR is used as the basis for a rule of thumb for identifying outliers.

The 1.5(IQR) Criterion for Outliers An observation is considered a suspected outlier if it is:

- below $Q1 - 1.5(IQR)$ or
- above $Q3 + 1.5(IQR)$

The following picture illustrates this rule:

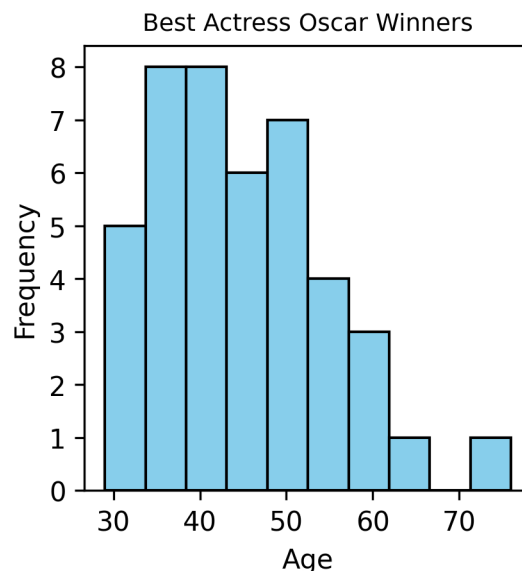


🔗 Example (C) Best Actress Oscar Winners

We will continue with the Best Actress Oscar winners example.

34 34 27 37 42 41 36 32 41 33 31 74 33 49 38 61 21 41 26 80 42 29 33 36 45 49 39 34 26 25 33 35 35 28 30
29 61 32 33 45 29 62 22 44

Recall that when we first looked at the histogram of ages of Best Actress Oscar winners, there were 5 observations that looked like possible outliers:



We can now use the $1.5(IQR)$ criterion to check whether the 5 observations should indeed be classified as outliers:

- For this example we found that $Q1 = 30.5$ and $Q3 = 42.5 \Rightarrow IQR = 11.5$.
- $Q1 - 1.5(IQR) = 30.5 - (1.5)(11.5) = 13.25$.
- $Q3 + 1.5(IQR) = 42.5 + (1.5)(11.5) = 59.25$.

The $1.5(IQR)$ criterion tells us that any observation that is below 13.25 or above 59.25 is considered a suspected outlier.

We therefore conclude that the observations 61, 61, 62, 74 and 80 should be flagged as suspected outliers in the distribution of ages. Note that since the smallest observation is 21, there are no suspected low outliers in this distribution.

4.4.9 Module 4 | Measures of Spread (5 of 5)

Learning Objectives

- Summarize and describe the distribution of a quantitative variable in context: a) describe the overall pattern, b) describe striking deviations from the pattern.
- Relate measures of center and spread to the shape of the distribution, and choose the appropriate measures in different contexts.

Understanding Outliers We just practiced one way to ‘flag’ possible outliers. Why is it important to identify possible outliers, and how should they be dealt with? The answers to these questions depend on the reasons for the outlying values. Here are several possibilities:

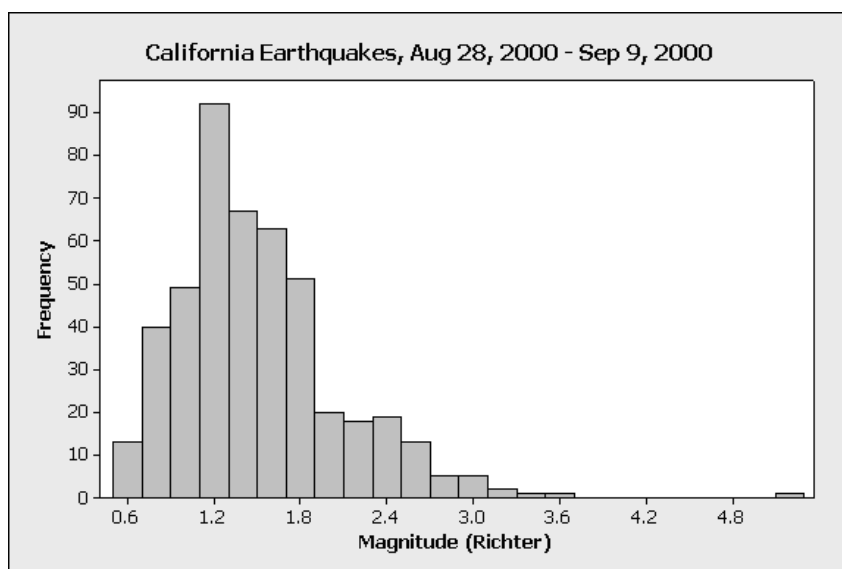
1. Even though it is an extreme value, if an outlier can be understood to have been produced by *essentially the same sort of physical or biological process* as the rest of the data, and if such extreme values are expected to

eventually occur again, then such an outlier indicates something important and interesting about the process you're investigating, and it *should be kept* in the data.

2. If an outlier can be explained to have been produced under fundamentally *different* conditions from the rest of the data (or by a fundamentally different process), such an outlier *can be removed* from the data if your goal is to investigate only the process that produced the rest of the data.
3. An outlier might indicate a *mistake* in the data (like a typo, or a measuring error), in which case it *should be corrected if possible or else removed* from the data before calculating summary statistics or making inferences from the data (and the reason for the mistake should be investigated).

Here are examples of each of these types of outliers:

1. The following histogram displays the magnitude of 460 earthquakes in California, occurring in the year 2000, between August 28 and September 9:



Identifying the outlier:

On the very far right edge of the display (beyond 4.8), we see a low bar; this represents one earthquake (because the bar has height of 1) that was much more severe than the others in the data.

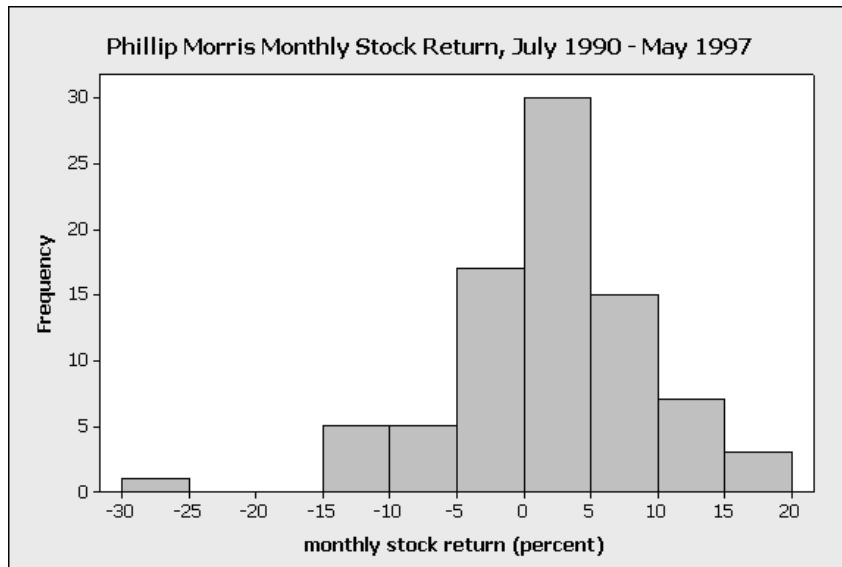
Understanding the outlier:

In this case, the outlier represents a much stronger earthquake, which is relatively rarer than the smaller quakes that happen more frequently in California.

How to handle the outlier:

For many purposes, the relatively severe quakes represented by the outlier might be the most important (because, for instance, that sort of quake has the potential to do more damage to people and infrastructure). The smaller-magnitude quakes might not do any damage, or even be felt at all. So, for many purposes it could be important to keep this outlier in the data.

2. The following histogram displays the monthly percent return on the stock of Phillip Morris (a large tobacco company) from July 1990 to May 1997:



Identifying the outlier:

On the display, we see a low bar far to the left of the others; this represents one month's return (because the bar has height of 1), where the value of Phillip Morris stock was unusually low.

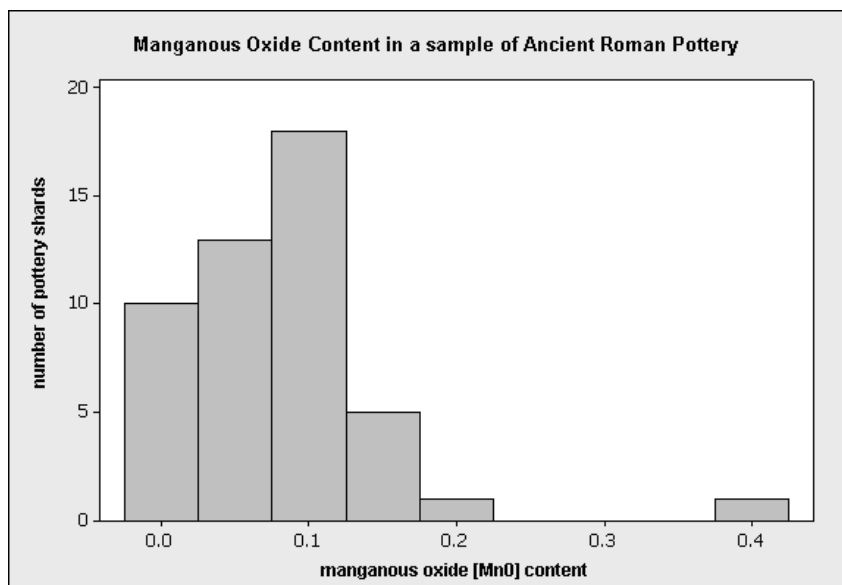
Understanding the outlier:

The explanation for this particular outlier is that, in the early 1990s, there were highly-publicized federal hearings being conducted regarding the addictiveness of smoking, and there was growing public sentiment against the tobacco companies. The unusually low monthly value in the Phillip Morris dataset was due to public pressure against smoking, which negatively affected the company's stock for that particular month.

How to handle the outlier:

In this case, the outlier was due to unusual conditions during one particular month that aren't expected to be repeated, and that were fundamentally different from the conditions that produced the values in all the other months. So in this case, it would be reasonable to remove the outlier, if we wanted to characterize the 'typical' monthly return on Phillip Morris stock.

- When archaeologists dig up objects such as pieces of ancient pottery, chemical analysis can be performed on the artifacts. The chemical content of pottery can vary depending on the type of clay as well as the particular manufacturing technique. The following histogram displays the results of one such actual chemical analysis, performed on 48 ancient Roman pottery artifacts from archaeological sites in Britain:



Identifying the outlier:

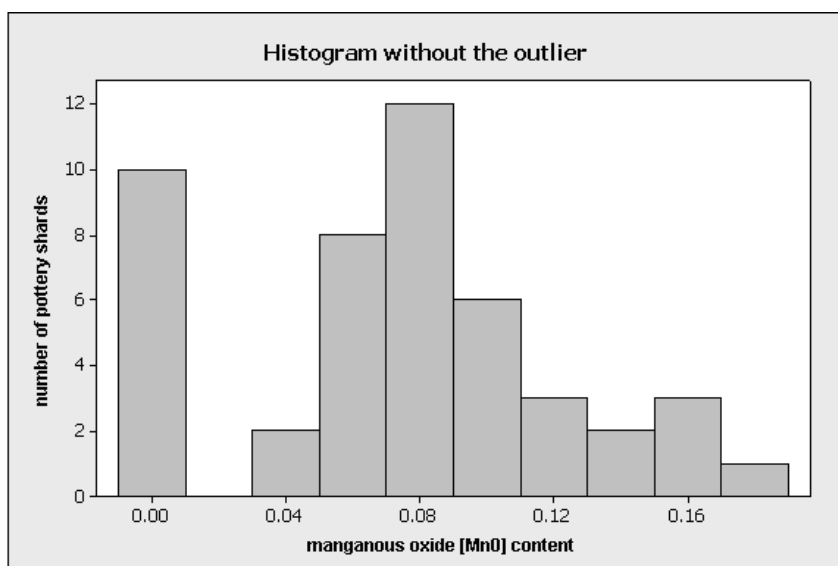
On the display, we see a low bar far to the right of the others; this represents one piece of pottery (because the bar has a height of 1), which has a suspiciously high manganous oxide value.

Understanding the outlier:

Based on comparison with other pieces of pottery found at the same site, and based on expert understanding of the typical content of this particular compound, it was concluded that the unusually high value was most likely a typo that was made when the data were published in the original 1980 paper (it was typed as “.394” but it was probably meant to be “.094”).

How to handle the outlier:

In this case, since the outlier was judged to be a mistake, it should be removed from the data before further analysis. In fact, removing the outlier is useful not only because it's a mistake, but also because doing so reveals important structure that was otherwise hidden. This feature is evident on the next display:



When the outlier is removed, the display is re-scaled so that now we can see the set of 10 pottery pieces that had almost no manganous oxide. These 10 pieces might have been made with a different potting technique, so identifying them as different from the rest is historically useful. This feature was only evident after the outlier was removed.

4.4.10 Let's Summarize

- The range covered by the data is the most intuitive measure of spread and is exactly the distance between the smallest data point (min) and the largest one (Max).
- Another measure of spread is the inter-quartile range (IQR), which is the range covered by the middle 50% of the data.
- $IQR = Q3 - Q1$, the difference between the third and first quartiles. The first quartile ($Q1$) is the value such that one quarter (25%) of the data points fall below it, or the median of the bottom half of the data. The third quartile is the value such that three quarters (75%) of the data points fall below it, or the median of the top half of the data.
- The IQR should be used as a measure of spread of a distribution only when the median is used as a measure of center.
- The IQR can be used to detect outliers using the $1.5(IQR)$ criterion. Outliers are observations that fall below $Q1 - 1.5(IQR)$ or above $Q3 + 1.5(IQR)$.

4.4.11 Module 4 | Boxplot (1 of 3)

Learning Objectives

- Generate and interpret several different graphical displays of the distribution of a quantitative variable (histogram, stemplot, boxplot).

Introduction Before we move on to the third measure of spread (standard deviation), we'll summarize what we've learned so far about measuring spread and use it to introduce another graphical display of the distribution of a quantitative variable, the *boxplot*.

The Five Number Summary So far, in our discussion about measures of spread, the key players were:

- the extremes (min and Max), which provide the range covered by all the data; and
- the quartiles (Q1, M and Q3), which together provide the IQR, the range covered by the middle 50% of the data.

The combination of all five numbers (min, Q1, M , Q3, Max) is called the *five number summary*, and provides a quick numerical description of both the center and spread of a distribution.

Example (D) Best Actress Oscar Winners

We will continue with the Best Actress Oscar winners example.

34 34 27 37 42 41 36 32 41 33 31 74 33 49 38 61 21 41 26 80 42 29 33 36 45 49 39 34 26 25 33 35 35 28 30
29 61 32 33 45 29 62 22 44

The five number summary of the age of Best Actress Oscar winners (1970-2013) is:

Min: 21

Q1: 30.5

M : 34.5

Q3: 42

Max: 80

4.4.12 Module 4 | Boxplot (2 of 3)

Learning Objectives

- Generate and interpret several different graphical displays of the distribution of a quantitative variable (histogram, stemplot, boxplot).

Now that you understand what each of the five numbers means, you can appreciate how much information about the distribution is packed into the five-number summary. All this information can also be represented visually by using the boxplot.

The Boxplot The boxplot graphically represents the distribution of a quantitative variable by visually displaying the five-number summary and any observation that was classified as a suspected outlier using the $1.5(\text{IQR})$ criterion.

There are several ways to plot the whiskers on a boxplot. One convention is to plot whiskers down to the minimum and up to the maximum value. We use the $1.5(\text{IQR})$ criterion, also known as the Tukey method for plotting whiskers. First, calculate the IQR, the difference between the 75th and 25th percentiles (or $Q3 - Q1$). Multiply the IQR by 1.5. Add this value to the 75th percentile. If the value is greater than (or equal to) the maximum value in the dataset, draw the upper whisker to the maximum value. Otherwise, stop the whisker at the largest value that is less than 75th percentile + $1.5 \times \text{IQR}$. Plot any values that are greater than this as individual points that are outliers. Similarly, subtract $1.5 \times \text{IQR}$ from the 25th percentile. If this value is smaller than the minimum value in the dataset, draw the lower whisker to the minimum value. Otherwise, stop the whisker at the lowest value that is greater than 25th percentile - $1.5 \times \text{IQR}$. Plot any values that are smaller than this as individual points that are outliers.

Using the Best Actress dataset, here is how we determine where to draw the whiskers:

- $Q3 = 42$
- $Q1 = 30.5$
- $\text{IQR}: 42 - 30.5 = 11.5$

- $1.5 \times \text{IQR} = 1.5 \times 11.5 = 17.25$
- $Q3 + 1.5 \times \text{IQR} = 42 + 17.25 = 59.25$

The largest observation that is less than or equal to 59.25 is 49 so we draw the upper whisker up to 49. All points above 49 are considered outliers (61, 61, 62, 74, 80).

$$Q1 - 1.5 \times \text{IQR} = 30.5 - 17.25 = 13.25$$

The smallest observation that is greater than or equal to 13.5 is 21 so we draw the lower whisker down to 21, which is also the minimum. There are no outliers.

To see how a boxplot is constructed see the video here: <https://www.youtube.com/watch?v=S50-WYp0m4I> (this is for the “Best Actress” dataset).

4.4.13 Module 4 | Boxplot (3 of 3)

Learning Objectives

- Compare and contrast distributions (of quantitative data) from two or more groups, and produce a brief summary, interpreting your findings in context.

Side-By-Side (Comparative) Boxplots As we learned in the beginning of this module, the distribution of a quantitative variable is best represented graphically by a histogram. Boxplots are most useful when presented side-by-side for comparing and contrasting distributions from two or more groups.

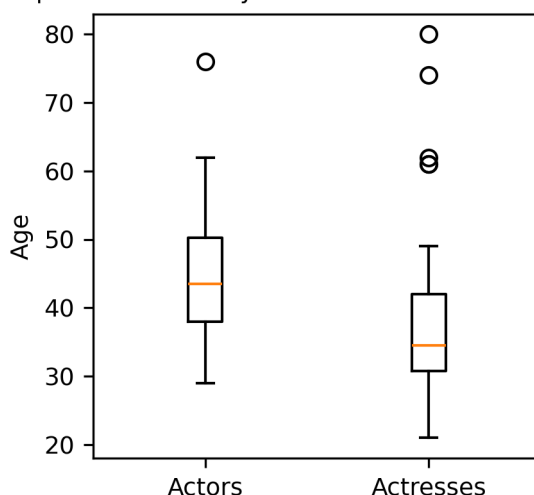
Example (E) Best Actor/Actress Oscar Winners

So far we have examined the age distributions of Oscar winners for males and females separately.

It will be interesting to *compare* the age distributions of actors and actresses who won best acting Oscars. To do that we will look at side-by-side boxplots of the age distributions by gender. Recall also that we found the five-number summary and means for both distributions. For the Best Actress dataset, we did the calculations by hand. For the Best Actor dataset, we used statistical software, and here are the results we got:

- Actors: min = 31, $Q1 = 38$, $M = 43.5$, $Q3 = 50.5$, Max = 76
- Actresses: min = 21, $Q1 = 30.5$, $M = 34.5$, $Q3 = 42$, Max = 80

Boxplots of Best Hollywood Actors and Actresses Data



Based on the graph and numerical measures, we can make the following comparison between the two distributions:

Center: The graph reveals that the age distribution of the males is higher than the females’ age distribution. This is supported by the numerical measures. The median age for females (34.5) is lower than for the males (43.5).

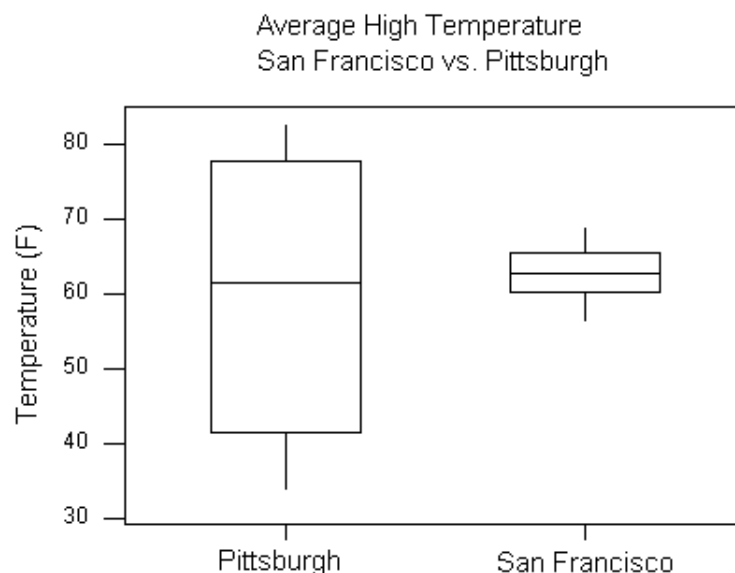
Actually, it should be noted that even the third quartile of the females' distribution (42) is lower than the median age for males. We therefore conclude that in general, actresses win the Best Actress Oscar at a younger age than actors do.

Spread: Judging by the range of the data, there is much more variability in the females' distribution (range = 59) than there is in the males' distribution (range = 47). On the other hand, if we look at the IQR, which measures the variability only among the middle 50% of the distribution, we see slightly more spread in the ages of males (IQR = 12.5) than females (IQR = 11.5). We conclude that among all the winners, the actors' ages are more alike than the actresses' ages. However, the middle 50% of the age distribution of actresses is more homogeneous than the actors' age distribution.

Outliers: We see that we have outliers in both distributions. There is only one high outlier in the actors' distribution (76, Henry Fonda, *On Golden Pond*), compared with three high outliers in the actresses' distribution.

Example (F) Temperature of Pittsburgh vs. San Francisco

In order to compare the average high temperatures of Pittsburgh to those in San Francisco we will look at the following side-by-side boxplots, and supplement the graph with the descriptive statistics of each of the two distributions.



Statistic	San Francisco	Pittsburgh
min	56.3	33.7
Q1	60.2	41.2
Median	62.7	61.4
Q3	65.35	77.75
Max	68.7	82.6

When looking at the graph, the similarities and differences between the two distributions are striking. Both distributions have roughly the same center (medians are 61.4 for Pitt, and 62.7 for San Francisco). However, the temperatures in Pittsburgh have a much larger variability than the temperatures in San Francisco (Range: 49 vs. 12. IQR: 36.5 vs. 5).

The practical interpretation of the results we got is that the weather in San Francisco is much more consistent than the weather in Pittsburgh, which varies a lot during the year. Also, because the temperatures in San Francisco vary so little during the year, knowing that the median temperature is around 63 is actually very informative. On the other hand, knowing that the median temperature in Pittsburgh is around 61 is practically useless, since temperatures vary so much during the year, and can get much warmer or much colder.

Note that this example provides more intuition about variability by interpreting small variability as consistency, and large variability as lack of consistency. Also, through this example we learned that the center of the distribution is more meaningful as a typical value for the distribution when there is little variability (or, as statisticians say, little “noise”) around it. When there is large variability, the center loses its practical meaning as a typical value.

4.4.14 Let's Summarize

- The five-number summary of a distribution consists of the median (M), the two quartiles (Q1, Q3) and the extremes (min, Max).
- The five-number summary provides a complete numerical description of a distribution. The median describes the center, and the extremes (which give the range) and the quartiles (which give the IQR) describe the spread.
- The boxplot graphically represents the distribution of a quantitative variable by visually displaying the five number summary and any observation that was classified as a suspected outlier using the 1.5 (IQR) criterion.
- Boxplots are most useful when presented side-by-side to compare and contrast distributions from two or more groups.

4.4.15 Module 4 | Standard Deviation (1 of 3)

Learning Objectives

- Relate measures of center and spread to the shape of the distribution, and choose the appropriate measures in different contexts.

Introduction So far, we have introduced two measures of spread; the range (covered by all the data) and the inter-quartile range (IQR), which looks at the range covered by the middle 50% of the distribution. We also noted that the IQR should be paired as a measure of spread with the median as a measure of center. We now move on to another measure of spread, the *standard deviation*, which quantifies the spread of a distribution in a completely different way.

Idea The idea behind the standard deviation is to quantify the spread of a distribution by measuring how far the observations are from their mean, $\bar{x}$. The standard deviation gives the average (or typical distance) between a data point and the mean, $\bar{x}$.

Notation There are many notations for the standard deviation: SD, s, Sd, StDev. Here, we'll use *SD* as an abbreviation for standard deviation, and use *s* as the symbol.

Calculation In order to get a better understanding of the standard deviation, it would be useful to see an example of how it is calculated. In practice, we will use a computer to do the calculation.

Example (G) Video Store Customers

The following are the number of customers who entered a video store in 8 consecutive hours: 7, 9, 5, 13, 3, 11, 15, 9

To find the standard deviation of the number of hourly customers:

1. Find the mean, $\bar{x}$ of your data: $\frac{7 + 9 + 5 + \dots + 9}{8} = 9$
2. Find the deviations from the mean: the difference between each observation and the mean
 $(7 - 9), (9 - 9), (5 - 9), (13 - 9), (3 - 9), (11 - 9), (15 - 9), (9 - 9)$
 $-2, 0, -4, 4, -6, 2, 6, 0$

Since the standard deviation is the average (typical) distance between the data points and their mean, it would make sense to average the deviations we got. Note, however, that the sum of the deviations from the

mean, $\bar{x}$ is 0 (add them up and see for yourself). This is always the case, and is the reason why we have to do a more complicated calculation to determine the standard deviation:

3. Square each of the deviations:

The first few are

$(-2)^2 = 4$, $(0)^2 = 0$, $(-4)^2 = 16$, and the rest are 16, 36, 4, 36, 0.

4. Average the square deviations by adding them up, and dividing by $n - 1$, (one less than the sample size):

$$\frac{4 + 0 + 16 + 36 + 4 + 36 + 0}{8 - 1} = \frac{112}{7} = 16.$$

- the reason why we “sort of” average the square deviations (divide by $n - 1$) rather than take the actual average (divide by n) is beyond the scope of the course at this point.
- This average of the squared deviations is called the *variance* of the data.

5. The SD of the data is the square root of the variance: $SD = \sqrt{16} = 4$.

- Why do we take the square root? Note that 16 is an average of the squared deviations, and therefore has different units of measurement. In this case 16 is measured in “squared customers,” which obviously cannot be interpreted. We therefore take the square root in order to compensate for the fact that we squared our deviations, and in order to go back to the original unit of measurement.

Recall that the average number of customers who enter the store in an hour is 9. The interpretation of $SD = 4$ is that on average, the actual number of customers that enter the store each hour is 4 away from 9.

Comment The importance of the numerical figure that we found in #4 above called the variance (=16 in our example) may be discussed in a course that covers inference.

Properties of the Standard Deviation

1. It should be clear from the discussion thus far that the SD should be paired as a measure of spread with the mean as a measure of center.
2. Note that the only way, mathematically, in which the $SD = 0$, is when all the observations have the same value (Ex: 5, 5, 5, ... , 5), in which case, the deviations from the mean (which is also 5) are all 0. This is intuitive, since if all the data points have the same value, we have no variability (spread) in the data, and expect the measure of spread (like the SD) to be 0. Indeed, in this case, not only is the SD equal to 0, but the range and the IQR are also equal to 0. Do you understand why?
3. Like the mean, the SD is strongly influenced by outliers in the data. Consider the example concerning video store customers: 3, 5, 7, 9, 9, 11, 13, 15 (data ordered). If the largest observation was wrongly recorded as 150, then the average would jump up to $\bar{x} = 25.9$, and the standard deviation would jump up to $SD = 50.3$. Note that in this simple example, it is easy to see that while the standard deviation is strongly influenced by outliers, the IQR is not. The IQR would be the same in both cases, since, like the median, the calculation of the quartiles depends only on the order of the data rather than the actual values.

The last comment leads to the following very important conclusion:

Choosing Numerical Summaries Use $\bar{x}$ (the mean) and the standard deviation as measures of center and spread *only* for reasonably symmetric distributions with no outliers.

Use the five-number summary (which gives the median, IQR and range) for all other cases.

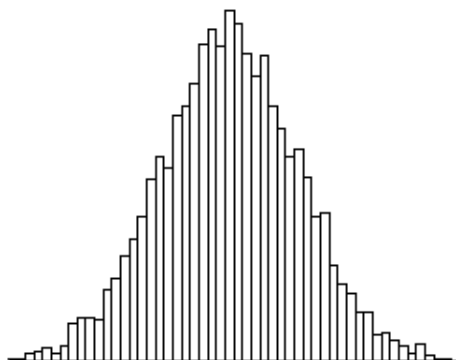
4.4.16 Module 4 | Standard Deviation (2 of 3)

Learning Objectives

- Apply the standard deviation rule to the special case of distributions having the “normal” shape.

The Standard Deviation Rule In the previous activity we tried to help you develop better intuition about the concept of standard deviation. The rule that we are about to present, called “The Standard Deviation Rule” (also known as “The Empirical Rule”) will hopefully also contribute to building your intuition about this concept.

Consider a symmetric mound-shaped distribution:

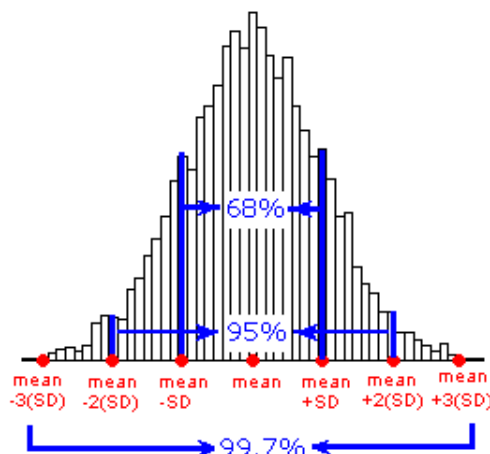


For distributions having this shape (also known as the *normal* shape), the following rule applies:

The Standard Deviation Rule:

- Approximately 68% of the observations fall within 1 standard deviation of the mean.
- Approximately 95% of the observations fall within 2 standard deviations of the mean.
- Approximately 99.7% (or virtually all) of the observations fall within 3 standard deviations of the mean.

The following picture illustrates this rule:



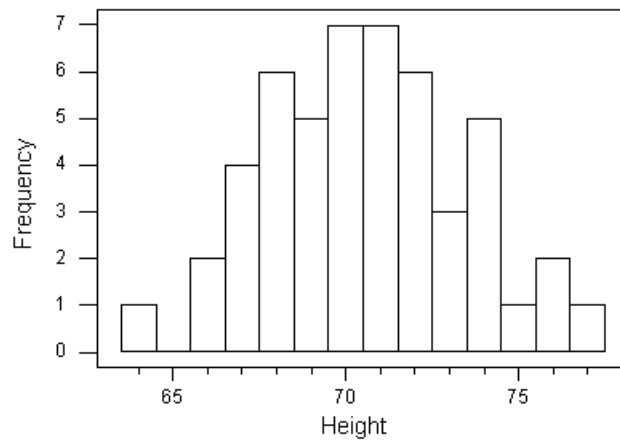
This rule provides another way to interpret the standard deviation of a distribution, and thus also provides a bit more intuition about it.

To see how this rule works in practice, consider the following example:

Example (H) Male Height

The following histogram represents height (in inches) of 50 males. Note that the data are roughly normal, so we would like to see how the Standard Deviation Rule works for this example.

ctd. on next page



Below are the actual data, and the numerical summaries of the distribution. Note that the key players here, the mean and standard deviation, have been highlighted.

64 66 66 67 67 67 67 68 68 68
 68 68 68 69 69 69 69 69 70 70
 70 70 70 70 70 71 71 71 71 71
 71 71 72 72 72 72 72 72 73 73
 73 74 74 74 74 74 75 76 76 77

Statistic	Height
N	50
Mean	70.58
StDev	2.858
min	64
Q1	68
Median	70.5
Q3	72
Max	77

To see how well the Standard Deviation Rule works for this case, we will find what percentage of the observations falls within 1, 2, and 3 standard deviations from the mean, and compare it to what the Standard Deviation Rule tells us this percentage should be.

Interval	Mean-SD, Mean+SD (67.7 , 73.4)	Mean-2(SD), Mean+2(SD) (64.9 , 76.3)	Mean-3(SD), Mean+3(SD) (62 , 79.2)
Percentage of Observations in interval	34 observations 34/50 = 68%	48 observations 48/50 = 96%	All 50 observations 50/50 = 100%
SD Rule says...	68%	95%	99.7%

It turns out the Standard Deviation Rule works *very well* in this example.

4.4.17 Module 4 | Standard Deviation (3 of 3)

Learning Objectives

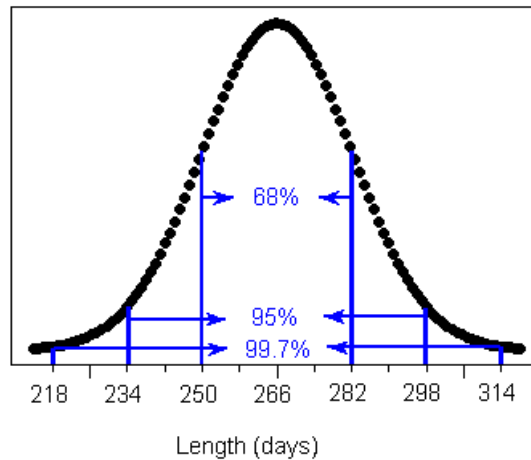
- Apply the standard deviation rule to the special case of distributions having the “normal” shape.

The following example illustrates how we can apply the Standard Deviation Rule to variables whose distribution is known to be approximately normal.

Example (I) Length of Human Pregnancy

The length of the human pregnancy is not fixed. It is known that it varies according to a distribution which is roughly normal, with a mean of 266 days, and a standard deviation of 16 days. (Source: Figures are from Moore and McCabe, *Introduction to the Practice of Statistics*).

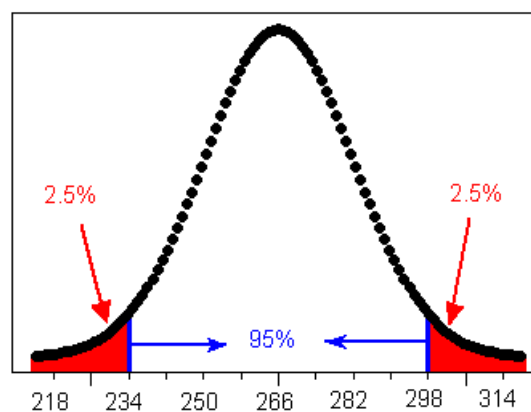
First, let's apply the Standard Deviation Rule to this case by drawing a picture:



We can now use the information provided by the Standard Deviation Rule about the distribution of the length of human pregnancy, to answer some questions. For example:

How long do the middle 95% of human pregnancies last? The middle 95% of pregnancies last within 2 standard deviations of the mean, or in this case 234-298 days.

What percent of pregnancies last more than 298 days? To answer this consider the following picture:



Since 95% of the pregnancies last between 234 and 298 days, the remaining 5% of pregnancies last either less than 234 days or more than 298 days. Since the normal distribution is symmetric, these 5% of pregnancies are divided evenly between the two tails, and therefore 2.5% of pregnancies last more than 298 days.

How short are the shortest 2.5% of pregnancies? Using the same reasoning as in the previous question, the shortest 2.5% of human pregnancies last less than 234 days.

What percentage of human pregnancies last more than 266 days? Since 266 days is the mean, approximately 50% of pregnancies last more than 266 days.

In general, the larger the animal, the longer the length of pregnancy (also called *gestation period*). For the horse, for example, the gestation period varies roughly according to a normal distribution with a mean of 336 days and a standard deviation of 3 days (Source: These figures are from Moore and McCabe, *Introduction to the Practice of Statistics*).

4.4.18 Let's Summarize

- The standard deviation measures the spread by reporting a typical (average) distance between the data points and their average.
- It is appropriate to use the SD as a measure of spread with the mean as the measure of center.
- Since the mean and standard deviations are highly influenced by extreme observations, they should be used as numerical descriptions of the center and spread only for distributions that are roughly symmetric, and have no outliers.
- For symmetric mound-shaped distributions, the Standard Deviation Rule tells us what percentage of the observations falls within 1, 2, and 3 standard deviations of the mean, and thus provides another way to interpret the standard deviation's value for distributions of this type.

4.4.19 Module 4 | Wrap-Up (Distributions)

You have now completed the module on examining distributions, which is the first of two modules within the EDA unit. You've seen how, by using graphs and numbers, we can bring to light all the information that is stored in the data about the variable of interest. Another important point is that graphs and numbers are powerful only when accompanied by their interpretation in the context of the problem.

5 Module 5 | Examining Relationships

5.1 Introduction

Learning Objectives

- Classify a data analysis situation (involving two variables) according to the “role-type classification,” and state the appropriate display and/or numerical measures that should be used in order to summarize the data.
- Compare and contrast distributions (of quantitative data) from two or more groups, and produce a brief summary, interpreting your findings in context.
- Produce a two-way table, and interpret the information stored in it about the association between two categorical variables by comparing conditional percentages.
- Graphically display the relationship between two quantitative variables and describe: a) the overall pattern, and b) striking deviations from the pattern.
- Interpret the value of the correlation coefficient, and be aware of its limitations as a numerical measure of the association between two quantitative variables.
- In the special case of linear relationship, use the least squares regression line as a summary of the overall pattern, and use it to make predictions.
- Recognize the distinction between association and causation, and identify potential lurking variables for explaining an observed relationship.
- Recognize and explain the phenomenon of Simpson's Paradox as it relates to interpreting the relationship between two variables.

5.1.1 Module 5 | Examining Relationships

So far we have dealt with data obtained from one variable (either categorical or quantitative), and learned how to describe the distribution of the variable using the appropriate visual displays and numerical measures. In this section, examining relationships, we will look at *two* variables at a time and, as the title suggests, explore the relationship between them using (as before) visual displays and numerical summaries. This module starts with a section that gives general definitions, lays out the framework for the whole module, and lists the different cases that the module will cover.

5.2 Role-Type Classification

5.2.1 Module 5 | The Role-Type Classification (1 of 2)

Learning Objectives

- Classify a data analysis situation (involving two variables) according to the “role-type classification,” and state the appropriate display and/or numerical measures that should be used in order to summarize the data.

While it is fundamentally important to know how to describe the distribution of a single variable, most studies pose research questions that involve exploring the relationship between *two* variables using the collected data.

Here are a few examples of such research questions with the two variables highlighted:

Examples

1. Is there a relationship between *gender* and *test scores* on a particular standardized test?

Other ways of phrasing the same research question include the following:

- Is performance on the test related to gender?
 - Is there a gender effect on test scores?
 - Are there differences in test scores between males and females?
2. How is the *number of calories* in a hot dog related to (or affected by) the *type of hot dog* (beef, meat or poultry)?
In other words, are there differences in the number of calories among the three types of hot dogs?
 3. Is there a relationship between the *type of light* a baby sleeps with (no light, night-light, lamp) and whether or not the child develops *nearsightedness*?
 4. Are the *smoking habits* of a person (yes, no) related to the person’s *gender*?
 5. How well can we predict a student’s freshman year *GPA* from his/her *SAT score*?
 6. What is the relationship between driver’s *age* and sign legibility *distance* (the maximum distance at which the driver can read a sign)?
 7. Is there a relationship between the *time* a person has practiced driving while having a learner’s permit and *whether or not this person passed the driving test*?
 8. Can you predict a person’s *favorite type of music* (classical, rock, jazz) on the basis of his/her *IQ level*?

In most studies involving two variables, each of the variables has a role. We distinguish between:

- The *explanatory* variable (also commonly referred to as the *independent variable*) – the variable that claims to explain, predict, or affect the response; and
- The *response* variable (also commonly referred to as the *dependent variable*) – the outcome of the study.

Typically, the explanatory (or independent) variable is denoted by X , while the response (or dependent) variable is denoted by Y .

Explanatory and Response Variables In this course, we use the terms *explanatory* and *response* variables instead of *independent* and *dependent* variables.

Now, let’s go back to some of the examples and classify the two relevant variables according to their roles in the study.

🔗 Example (J) 1

We want to explore whether the outcome of the study – the score on a test – is affected by the test-taker’s gender.
Therefore:

Gender is the *explanatory* variable

Test score is the *response* variable

🧩 Example (K) 3

In this study, we explore whether the nearsightedness of a person can be explained by the type of light that person slept with as a baby. Therefore:

Light type is the *explanatory* variable

Nearsightedness is the *response* variable

🧩 Example (L) 5

Here we are examining whether a student's SAT score is a good predictor for the student's GPA freshman year. Therefore:

SAT score is the *explanatory* variable

GPA of freshman year is the *response* variable

🧩 Example (M) 7

Here we are examining whether a person's outcome on the driving test (pass/fail) can be explained by the length of time this person has practiced driving prior to the test. Therefore:

Time is the *explanatory* variable

Driving test outcome is the *response* variable

Is the role classification of variables always clear? In other words, is it always clear which of the variables is the explanatory and which is the response?

Answer: No. There are studies in which the role classification is not really clear. This mainly happens in cases when both variables are categorical or both are quantitative. An example is a study that explores the relationship between students' SAT Math and SAT Verbal scores. In cases like this, any classification choice would be fine (as long as it is consistent throughout the analysis).

5.2.2 Module 5 | The Role-Type Classification (2 of 2)

Learning Objectives

- Classify a data analysis situation (involving two variables) according to the “role-type classification,” and state the appropriate display and/or numerical measures that should be used in order to summarize the data.

If we further classify each of the two relevant variables according to *type* (categorical or quantitative), we get the following four possibilities for *role-type classification*:

1. Categorical explanatory and quantitative response
2. Categorical explanatory and categorical response
3. Quantitative explanatory and quantitative response
4. Quantitative explanatory and categorical response

This role-type classification can be summarized and easily visualized in the following table (note that the explanatory variable is always listed first):

		Response	
		Categorical	Quantitative
Explanatory	Categorical	$C \rightarrow C$	$C \rightarrow Q$
	Quantitative	$Q \rightarrow C$	$Q \rightarrow Q$

This role-type classification serves as the infrastructure for this entire section. In each of the four cases, different statistical tools (displays and numerical measures) should be used to explore the relationship between the two variables. This suggests the following important principle:

Principle When confronted with a research question that involves exploring the relationship between two variables, the first and most crucial step is to determine which of the four cases represents the data structure of the problem. In other words, the first step should be classifying the two relevant variables according to their role and type, and only then can we determine what statistical tools should be used to analyze them.

Now let's go back to our eight examples and determine which of the four cases represents the data structure of each:

Example (N) 1

Gender is the *explanatory* variable, and it is *categorical*.

Test score is the *response* variable, and it is *quantitative*.

Therefore, this is an example of case $C \rightarrow Q$.

Example (O) 3

Light Type is the *explanatory* variable, and it is *categorical*.

Nearsightedness is the *response* variable, and it is *categorical*.

Therefore, this is an example of case $C \rightarrow C$.

Example (P) 5

SAT Score is the *explanatory* variable, and it is *quantitative*.

GPA of Freshman Year is the *response* variable, and it is *quantitative*.

Therefore, this is an example of case $Q \rightarrow Q$.

Example (Q) 7

Time is the *explanatory* variable, and it is *quantitative*.

Driving Test Outcome is the *response* variable, and it is *categorical*.

Therefore, this is an example of case $Q \rightarrow C$.

The remainder of this section on exploring relationships is guided by this role-type classification. In the next three parts, we elaborate on cases $C \rightarrow Q$, $C \rightarrow C$, and $Q \rightarrow Q$. More specifically, we will learn the appropriate statistical tools (visual display and numerical summaries) that will allow us to explore the relationship between the two variables in each of the cases. Case $Q \rightarrow C$ is *not* discussed in this course and is typically covered in more advanced courses. The section concludes with a discussion on causal relationships.

5.3 Module 5 | Case C→Q

5.3.1 Module 5 | Case C→Q (1 of 2)

Learning Objectives

- Compare and contrast distributions (of quantitative data) from two or more groups, and produce a brief summary, interpreting your findings in context.

Case C→Q: Categorical Explanatory Variable and Quantitative Response Variable Recall the role-type classification table for framing our discussion about the relationship between two variables:

		Response	
		Categorical	Quantitative
Explanatory	Categorical	C → C	C → Q
	Quantitative	Q → C	Q → Q

We are now ready to start with Case C→Q, exploring the relationship between two variables where the explanatory variable is categorical, and the response variable is quantitative. As you’ll discover, exploring relationships of this type is something we’ve already discussed in this course, but we didn’t frame the discussion this way.

🔗 Example (R) Hot Dogs

Background: People who are concerned about their health may prefer hot dogs that are low in calories. A study was conducted by a concerned health group in which 54 major hot dog brands were examined, and their calorie contents recorded. In addition, each brand was classified by type: beef, poultry, and meat (mostly pork and beef, but up to 15% poultry meat). The purpose of the study was to examine whether the *number of calories* a hot dog has is related to (or affected by) its *type*.

Reference: Moore, David S., and George P. McCabe (1989). *Introduction to the Practice of Statistics*. Original source: *Consumer Reports*, June 1986, pp. 366-367.

Answering this question requires us to examine the relationship between the categorical variable, Type and the quantitative variable Calories. Because the question of interest is whether the type of hot dog affects calorie content,

- the *explanatory* variable is *Type*, and
- the *response* variable is *Calories*.

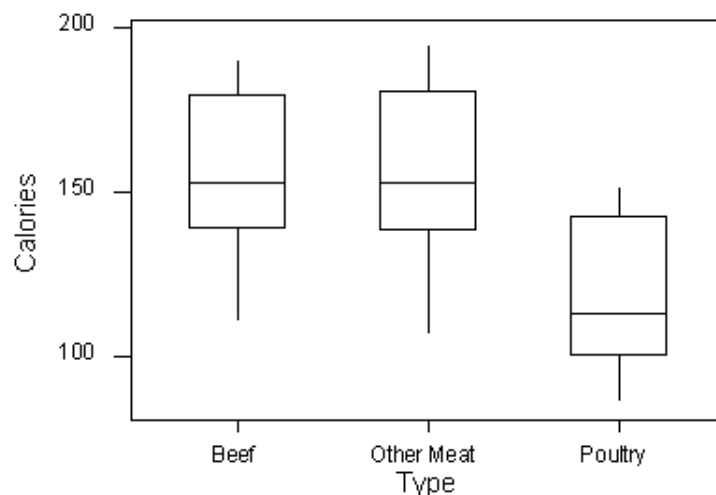
Here is what the raw data look like:

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	Explanatory	Response
	Type	Calories
Brand 1	Beef	186
Brand 2	Poultry	129
Brand 3	Beef	181
Brand 4	Meat	173
.	.	.
.	.	.
.	.	.
Brand 54	Poultry	144

The raw data are a list of types and calorie contents, and are not very useful in that form. To explore how the number of calories is related to the type of hot dog, we need an informative visual display of the data that will compare the three types of hot dogs with respect to their calorie content.

The visual display that we'll use is *side-by-side boxplots* (which we've seen before). The side-by-side boxplots will allow us to *compare the distribution* of calorie counts within each category of the explanatory variable, hot dog type:



As before, we supplement the side-by-side boxplots with the descriptive statistics of the calorie content (response) for each type of hot dog separately (i.e., for each level of the explanatory variable separately):

Statistic	Beef	Meat	Poultry
min	111	107	86
Q1	139.5	138.5	100.5
Median	152.5	153	113
Q3	179.75	180.5	142.5
Max	190	195	152

Let's summarize the results we got and interpret them in the context of the question we posed:

By examining the three side-by-side boxplots and the numerical summaries, we see at once that poultry hot dogs, as a group, contain fewer calories than those made of beef or meat. The median number of calories in poultry hot dogs (113) is less than the median (and even the first quartile) of either of the other two distributions (medians 152.5 and 153). The spread of the three distributions is about the same, if IQR is considered (all slightly above 40), but the (full) ranges vary slightly more (beef: 80, meat:88, poultry: 66). The general recommendation to the health-conscious consumer is to eat poultry hot dogs. It should be noted, though, that since each of the three types of hot dogs shows quite a large spread among brands, simply buying a poultry hot dog does not guarantee a low-calorie food.

What we learn from this example is that when exploring the relationship between a categorical explanatory variable and a quantitative response (Case $C \rightarrow Q$), we essentially *compare the distributions of the quantitative response for each category of the explanatory variable* using side-by-side boxplots supplemented by descriptive statistics. Recall that we have actually done this before when we talked about the boxplot and argued that boxplots are most useful when presented side by side for comparing distributions of two or more groups. This is exactly what we are doing here!

5.3.2 Module 5 | Case $C \rightarrow Q$ (2 of 2)

Learning Objectives

- Compare and contrast distributions (of quantitative data) from two or more groups, and produce a brief summary, interpreting your findings in context.

Here is another example:

Example (S) SSHA

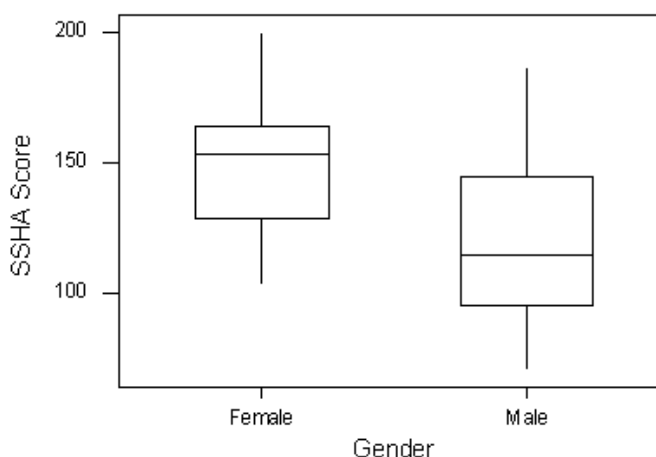
The Survey of Study Habits and Attitudes (SSHA) is a psychological test designed to measure the motivation, study habits, and attitudes toward learning of college students. Is there a relationship between *gender* and *SSHA scores*? In other words, is there a "gender effect" on SSHA scores? Data were collected from 40 randomly selected college students, and here is what the raw data look like:

	Explanatory	Response
	Gender	SSHA score
Student 1	Female	154
Student 2	Female	109
Student 3	Male	108
Student 4	Female	115
.	.	.
.	.	.
.	.	.
Student 40	Male	140

(Source: Moore, David S., and George P. McCabe. (2003). *Introduction to the Practice of Statistics*, 4th ed. New York: W. H. Freeman.)

Side-by-side boxplots supplemented by descriptive statistics allow us to compare the distribution of SSHA scores within each category of the explanatory variable – gender:

Statistic	Female	Male
min	103	70
Q1	128.75	95
Median	153	114.5
Q3	163.75	144.5
Max	200	187



5.3.3 Let's Summarize

- The relationship between a categorical explanatory variable and a quantitative response variable is summarized using:
 - Data display*: Side-by-side boxplots
 - Numerical summaries*: Descriptive statistics
- Exploring the relationship between a categorical explanatory variable and a quantitative response variable amounts to comparing the distributions of the quantitative response for each category of the explanatory variable. In particular, we look at how the distribution of the response variable differs between the values of the explanatory variable.

5.4 Module 5 | Case $C \rightarrow C$

5.4.1 Module 5 | Case $C \rightarrow C$ (1 of 2)

Learning Objectives

- Produce a two-way table, and interpret the information stored in it about the association between two categorical variables by comparing conditional percentages.

Case $C \rightarrow C$: Two Categorical Variables Recall the role-type classification table for framing our discussion about the relationship between two variables:

		Response	
		Categorical	Quantitative
Explanatory	Categorical	$C \rightarrow C$	✓ $C \rightarrow Q$
	Quantitative	$Q \rightarrow C$	$Q \rightarrow Q$

We are done with case $C \rightarrow Q$, and will now move on to case $C \rightarrow C$, where we examine the relationship between two categorical variables.

Earlier in the course, (when we discussed the distribution of a *single* categorical variable) we examined the data obtained when a random sample of 1,200 U.S. college students were asked about their bodyimage (underweight, overweight, or about right.) We are now returning to this example, to address the following question:

If we had separated our sample of 1,200 U.S. college students by gender and looked at *males and females separately*, would we have found a similar distribution across body-image categories? More specifically, are men and women just as likely to think their weight is about right? Among those students who do not think their weight is about right, is there a difference between the genders in feelings about bodyimage?

Answering these questions requires us to *examine the relationship between two categorical variables*, gender and body-image. Because the question of interest is whether there is a gender effect on body image,

- the *explanatory* variable is g , and
- the *response* variable is b .

Here is what the raw data look like when we include the gender of each student:

Explanatory **Response**



Student	Gender	Body Image
.	.	.
.	.	.
student 25	M	overweight
student 26	M	about right
student 27	F	underweight
student 28	F	about right
student 29	M	about right
.	.	.
.	.	.

Once again the raw data is a long list of 1,200 genders and responses, and thus not very useful in that form. To start our exploration of how body image is related to gender, we need an informative display that summarizes the data. In order to summarize the relationship between two categorical variables, we create a display called a *two-way table*.

Here is the two-way table for our example:

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	560	163	37	760
	Male	295	72	73	440
	Total	855	235	110	1200

The table has the possible genders in the rows, and the possible responses regarding body image in the columns. At each intersection between row and column, we put the counts for how many times that combination of gender and body image occurred in the data. We sum across the rows to fill in the Total column, and we sum across the columns to fill in the Total row.

Comment Note that from the way the two-way table is constructed, the Total row or column is a summary of one of the two categorical variables, ignoring the other. In our example:

- The Total row gives the summary of the categorical variable bodyimage:

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	560	163	37	760
	Male	295	72	73	440
	Total	855	235	110	1200

(These are the same counts we got earlier in the course when we looked at the single categorical variable bodyimage, and did not consider gender.)

- The Total column gives the summary of the categorical variable gender:

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	560	163	37	760
	Male	295	72	73	440
	Total	855	235	110	1200

5.4.2 Module 5 | Case C→C (2 of 2)

Learning Objectives

- Produce a two-way table, and interpret the information stored in it about the association between two categorical variables by comparing conditional percentages.

So far, we have organized the raw data in a much more informative display – the two-way table:

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	560	163	37	760
	Male	295	72	73	440
	Total	855	235	110	1200

Remember, though, that our primary goal is to explore how body image is related to gender. Exploring the relationship between two categorical variables (in this case, body image and gender) amounts to comparing the distributions of the response variable (in this case body image) across the different values of the explanatory variable (in this case, males and females):

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	560	163	37	760
	Male	295	72	73	440
	Total	855	235	110	1200

Compare these distributions! →

Note that it doesn't make sense to compare raw counts, because there are more females than males overall. So, for example, it is not very informative to say, "There are 560 females who responded 'about right' compared to only 295 males," since the 560 females are out of a total of 760, and the 295 males are out of a total of only 440.

We need to supplement our display, the two-way table, with some numerical summaries that will allow us to compare the distributions. These numerical summaries are found by simply *converting the counts to percentages within (or restricted to) each value of the explanatory variable separately*.

In our example, we look at each gender separately and convert the counts to percentages *within that gender*. Let's start with females:

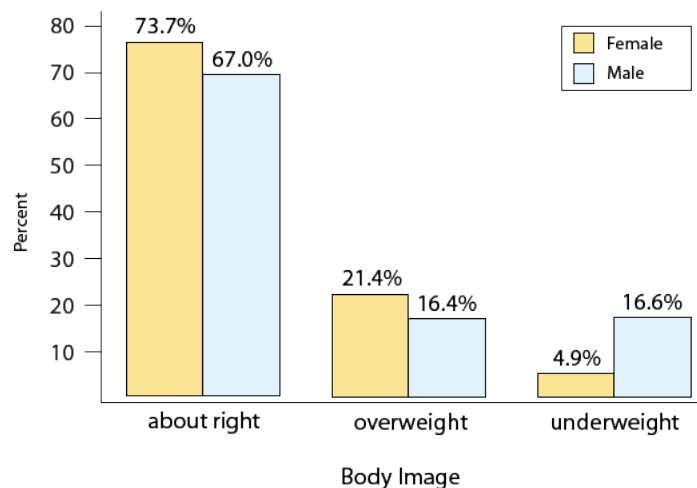
		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	$560/760 = 73.7\%$	$163/760 = 21.5\%$	$37/760 = 4.9\%$	$760/760 = 100\%$
	Male	%	%	%	%

Note that each count is converted to percents by dividing by the total number of females, 760. These numerical summaries are called *conditional percentages*, since we find them by “conditioning” on one of the genders.

Comments:

1. In our example, we chose to organize the data with the explanatory variable gender in rows and the response variable body image in columns, and thus our conditional percentages were *row percentages*, calculated within each row separately. Similarly, if the explanatory variable happens to sit in columns and the response variable in rows, our conditional percentages will be *column percentages*, calculated within each column separately. For an example, see the “Did I Get This?” exercises below.
2. Another way to visualize the conditional percentages, instead of in a table, is to use a *double bar chart*. This display is quite common in newspapers.

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	73.7%	21.4%	4.9%	100%
	Male	67.0%	16.4%	16.6%	100%



Now that we have summarized the relationship between the categorical variables gender and body image, let’s go back and interpret the results in the context of the questions that we posed.

5.4.3 Let’s Summarize

- The relationship between two categorical variables is summarized using
 - *Data display*: Two-way table, supplemented by

– *Numerical summaries:* Conditional percentages.

- Conditional percentages are calculated for each value of the explanatory variable separately. They can be row percentages if the explanatory variable “sits” in the rows, or column percentages if the explanatory variable “sits” in the columns.
- When we try to understand the relationship between two categorical variables, we compare the distributions of the response variable for values of the explanatory variable. In particular, we look at how the pattern of conditional percentages differs between the values of the explanatory variable.